Identifying the Dominant B-Cell Transcriptional Signatures in Systemic Lupus Erythematosus and Lupus Nephritis

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Introduction

Systemic lupus erythematosus (SLE) is a chronic multisystem autoimmune disease characterized by loss of immune tolerance, persistent inflammation, and production of pathogenic autoantibodies. Although SLE can affect multiple organs, its clinical manifestations and severity vary considerably between patients, reflecting complex interactions among genetic susceptibility, environmental factors, and dysregulated immune responses.[1]

A central feature of SLE is the loss of immune tolerance, resulting in the recognition of self-antigens as targets of the immune response. This promotes the production of autoantibodies against nuclear antigens, including double-stranded DNA and ribonucleoproteins. Autoantibody-antigen interactions generate immune complexes that activate complement and initiate inflammatory responses, contributing to tissue damage.[2] The clinical consequences of these processes vary substantially among patients, reflecting the molecular and immunological heterogeneity of SLE and complicating efforts to predict disease progression and organ involvement.

Renal involvement represents one of the most clinically consequential manifestations of this heterogeneity. Lupus nephritis (LN) is one of the most severe complications of SLE, developing in approximately 25-60% of patients during the course of disease.[3] Deposition of immune complexes within the kidney and subsequent inflammatory responses can cause progressive renal tissue injury and, in severe cases, lead to end-stage kidney disease.[4] However, despite the substantial clinical burden of LN, it remains difficult to predict which patients with SLE will develop renal involvement. Current assessment relies on clinical measurements, serological biomarkers, and renal biopsy, while commonly used indicators such as anti-double-stranded DNA antibodies and complement levels may not reliably capture individual

disease progression.[5] Furthermore, renal biopsy is invasive and unsuitable for frequent monitoring. Thus, defining the molecular differences associated with renal involvement may provide insight into the mechanisms underlying progression from SLE to LN and identify disease-associated signatures that are not captured by conventional clinical measures.[4,5]

Among the immune cell populations implicated in SLE, B cells are particularly relevant to the development of LN because they contribute to several processes involved in autoimmune inflammation. Loss of B-cell tolerance can permit autoreactive B cells to become activated and produce autoantibodies, contributing to immune-complex formation and subsequent renal inflammation.[6] In addition to antibody production, B cells function as antigen-presenting cells and secrete cytokines that regulate other immune populations, allowing them to amplify and sustain autoimmune responses.[3-7] These diverse functions make B-cell transcriptional activity a potentially informative indicator of the molecular changes accompanying lupus progression. Therefore, characterizing B-cell gene expression patterns across healthy individuals, patients with SLE, and patients with LN may therefore help identify transcriptional programs associated with disease and renal involvement in particular.

Single-cell RNA sequencing (scRNA-seq) provides an effective approach for investigating this cellular heterogeneity by measuring gene expression at single-cell resolution. Unlike bulk RNA sequencing, which averages transcriptional signals across many cell types, scRNA-seq enables individual immune populations to be identified and examined independently.[8] This distinction is particularly important in SLE, where changes in cellular composition and activation state can obscure disease-associated signals in bulk measurements.[1,8] By first identifying the B-cell compartment and then examining its transcriptional profile across disease states, scRNA-seq can help distinguish B cell-specific changes from broader changes

occurring throughout the immune system.[8,9]

The current study therefore investigates B-cell transcriptional profiles across three clinical states: healthy controls (HC), patients with SLE, and patients with SLE who developed LN. Differential expression analyses will compare HC versus SLE, SLE versus LN, and HC versus LN to identify shared and disease-state associated transcriptional changes. Overlap among these comparisons will be examined to distinguish broadly lupus-associated genes from changes associated more specifically with the SLE to LN transition, followed by pathway enrichment analysis to characterize the biological processes represented by the identified genes. By focusing on transcriptional changes within B cells, this study aims to identify candidate molecular signatures associated with lupus progression and provide a foundation for their validation as potential biomarkers or mechanistic targets in future studies.

Methods

2.1 Study Design

This study investigated transcriptional differences among healthy controls (HC), patients with systemic lupus erythematosus (SLE), and patients with lupus nephritis (LN) using publicly available single-cell RNA sequencing (scRNA-seq) data. Peripheral blood mononuclear cell (PBMC) transcriptomes from nine samples were analyzed, comprising three HC, three SLE samples, and three LN samples (Table 1). The HC group consisted of individuals classified as healthy in the originating dataset, while SLE samples represent patients sampled prior to therapeutic intervention, thereby providing a pre-treatment transcriptional profile without the potential confounding effects of treatment. The LN group comprised patients with established renal involvement associated with SLE. Clinical group assignments and diagnostic criteria were

based on the metadata and eligibility criteria reported for the respective source datasets.

The computational workflow consisted of quality control, normalization, highly variable gene identification, dimensionality reduction, batch correction, cell clustering, UMAP visualization, and cell-type annotation. Following annotation, B cells will be isolated for downstream differential gene expression and pathway enrichment analyses to identify transcriptional changes associated with SLE and progression to LN (Figure 1).

2.2 Public Datasets

Single-cell RNA sequencing data were obtained from the GEO database as described in Table 1. All samples were PBMCs sequenced using the Illumina NovaSeq 6000 platform. The datasets were selected because their combined sample groups provided the three disease states required to investigate transcriptional differences associated with SLE and LN.

Table 1. *Samples included in this study.*

Condition	Sample accessions	GEO dataset
Healthy control	GSM9198093–GSM9198095	GSE306378
SLE pre-treatment	GSM8207595, GSM8207603, GSM8207605	GSE263931
Lupus nephritis	GSM9198096–GSM9198098	GSE306378

2.3 Computational Environment

All analyses were performed in R version 4.4.3 (2025-02-28) using RStudio version 2025.05.1+513 on macOS. Data preprocessing, normalization, dimensionality reduction, and clustering were performed using Seurat, while Harmony was used for dataset integration and

batch-effect correction. SingleR and celldex were used for automated cell-type annotation. Data manipulation and visualization were performed using dplyr, ggplot2, patchwork, and EnhancedVolcano. Package versions are provided in Table 2.

Table 2. *Software packages used for scRNA-seq analysis.*

Package	**Version**	**Purpose**
Seurat	5.5.1	Data preprocessing, normalization, dimensionality reduction, and clustering
Harmony	2.0.5	Dataset integration and batch-effect correction
SingleR	2.8.0	Automated cell-type annotation
celldex	1.16.0	Reference dataset for cell-type annotation
ggplot2	4.0.3	Data visualization
dplyr	1.2.1	Data manipulation
patchwork	1.3.2	Figure assembly
EnhancedVolcano	1.24.0	Volcano plot generation

2.4 Data Preprocessing and Quality Control

Expression matrices were imported into R and processed using Seurat. Quality control was performed to remove low-quality cells and potential technical artifacts prior to downstream analysis. Cells with fewer than 1,500 total RNA counts or fewer than 500 detected genes were excluded because low transcript and gene detection can indicate empty droplets, damaged cells, or insufficiently captured RNA. Cells with more than 3,200 detected genes were also excluded to reduce the inclusion of potential doublets or multiplets, in which transcripts from multiple cells are captured within a single droplet. Mitochondrial transcript abundance was used as an additional indicator of cellular quality, with cells containing greater than or equal to 20%

mitochondrial transcripts excluded because elevated mitochondrial RNA is commonly associated with stressed, damaged, or dying cells. Following filtering, gene expression data were normalized using Seurat's LogNormalize method, and highly variable genes were identified for downstream dimensionality reduction and clustering. The distributions of these quality-control metrics before filtering were assessed using violin plots (Figure 2).

2.5 Data Integration and Dimensionality Reduction

Principal component analysis (PCA) was performed using the identified highly variable genes to reduce the dimensionality of the gene expression data. Because each cell is characterized by the expression of thousands of genes, PCA transforms these measurements into a smaller set of principal components (PCs), with each PC representing a combination of genes that captures a major source of transcriptional variation across cells. This reduces the complexity of the dataset while retaining the major patterns of variation needed to identify transcriptionally similar cells. The number of principal components retained for downstream analysis was determined using an elbow plot (Figures 3-4), which identifies the point at which additional components contribute relatively little variation. Because the samples originated from two independent GEO datasets, Harmony was subsequently applied to the PCA representation using sample identity as the grouping variable to reduce sample-associated technical variation. The resulting Harmony-corrected dimensions were used for downstream clustering and Uniform Manifold Approximation and Projection (UMAP) visualization (Figure 5).

2.6 Cell Clustering and Annotation

Cells were grouped according to transcriptional similarity using graph-based clustering performed on the Harmony-corrected dimensions. Clustering was evaluated at resolutions of 0.1

and 0.5 to assess the level of cellular structure captured at different clustering granularities. UMAP was subsequently used to visualize the transcriptional relationships between cells in two dimensions. Cell identities were then assigned using SingleR with the DatabaseImmuneCellExpressionData reference from the celldex package. Both broad and fine resolution SingleR annotations were evaluated and compared with the expression of canonical marker genes to assess the biological consistency of the assigned cell identities.

2.7 Projected Downstream B-cell Analysis

The present study was completed through cell-type annotation, establishing the major immune populations within the integrated PBMC dataset. The subsequent analyses are proposed as the next stage of the project and will focus specifically on the B-cells.

Following cell-type annotation, B cells will be isolated from the broader PBMC population for downstream analysis. Rather than initially distinguishing detailed B-cell subtypes, the first analysis will examine transcriptional differences across the overall B-cell compartment to determine whether B-cell gene expression varies between disease states. Differential gene expression (DEG) analysis will compare HC versus SLE, SLE versus LN, and HC versus LN, with each comparison identifying genes whose expression differs between the respective groups. These comparisons will allow broadly lupus associated changes to be distinguished from transcriptional changes that are more specifically associated with LN. Genes identified across multiple comparisons, as well as genes unique to individual comparisons, will then be examined to characterize shared and disease-state-specific expression patterns. Finally, pathway enrichment analysis will be performed on the identified DEGs to determine whether the observed changes converge on particular biological processes or molecular pathways. Together, these analyses will establish a B-cell transcriptional profile of lupus progression and generate

candidate genes and pathways for subsequent validation and investigation.

Results

Following quality control, the integrated dataset retained more than 70,000 high-quality PBMCs across the three experimental conditions. The distributions of detected genes, total RNA counts, and mitochondrial transcript abundance demonstrated that the retained cells met the predefined quality criteria across HC, SLE, and LN samples (Figure 2). These metrics indicated that the dataset contained sufficient transcriptomic information for downstream analysis while minimizing the contribution of low-quality or damaged cells. The retained cells were subsequently normalized and highly variable genes were identified to capture the features contributing most strongly to cellular heterogeneity.

Principal component analysis revealed substantial transcriptional variation across individual cells. The elbow plot showed a clear reduction in the contribution of successive principal components, with an inflection around PC15, supporting the use of the first 15 PCs for subsequent dimensionality reduction and clustering (Figure 3). Examination of the genes contributing to these components demonstrated that the observed variation was associated with recognizable immune-cell and functional programs rather than being uniformly distributed across genes (Figure 4). For example, PC1 showed strong contributions from myeloid-associated genes including FCN1, MNDA, CD14, and LYZ, consistent with variation associated with the monocyte compartment.[10] PC4 contained prominent B-cell associated genes including MS4A1, CD79A, and BANK1, indicating that B-cell identity contributed substantially to the transcriptional structure of the dataset. PC11 was characterized by interferon-stimulated genes including ISG15 and IFI6, while later components included genes associated with

antibody-secreting cell differentiation, including JCHAIN, MZB1, and IGHA1.[11] Thus, the retained PCs captured distinct biological programs corresponding to major immune-cell populations and disease-relevant transcriptional activity.

Because the samples originated from two independent GEO datasets, sample-associated variation was assessed during integration. Before correction, cellular distributions showed evidence of sample-specific separation, indicating that technical variation could influence the transcriptional structure of the combined dataset. Harmony integration using sample identity reduced this separation and produced greater mixing of cells from the nine samples within shared transcriptional populations (Figure 5). This improved integration was important for subsequent comparisons because it reduced the likelihood that clustering reflected dataset or patient-specific technical effects alone. The Harmony-corrected dimensions were then used for graph-based clustering, which identified 21 transcriptionally distinct clusters, spatially separating populations according to similarities in their gene-expression profiles (Figure 6).

Automated annotation using SingleR and the celldex immune reference dataset assigned the 21 clusters to major immune populations. Broad annotation identified 25,483 NK cells, 18,803 monocytes, 17,678 CD4+ T cells, 11,099 CD8+ T cells, and 3,353 B cells (Figure 7). Thus, B cells represented a smaller but clearly identifiable population within the overall PBMC dataset. Fine-resolution annotation further classified 3,340 of the B cells as naive B cells, indicating that the majority of the identified B-cell population shared a transcriptional profile consistent with a naive state. This assignment was supported by marker expression, with TCL1A and IGHD showing concentrated expression within the corresponding B-cell population (Figure 4). The agreement between automated annotation and canonical marker expression provided independent support for the identification of the B-cell population and established it as a suitable

target for subsequent disease-associated transcriptional analysis.

Discussion

The present study established an integrated single-cell transcriptomic map of the peripheral immune compartment across healthy controls (HC), systemic lupus erythematosus (SLE), and lupus nephritis (LN), providing a framework for investigating disease associated transcriptional changes at a cellular resolution. Principal component analysis (PCA) revealed biologically interpretable sources of transcriptional variation across the dataset, including B-cell associated genes such as *MS4A1*, *CD79A*, and *BANK1*, interferon-stimulated genes including *ISG15* and *IFI6*, and genes associated with antibody-secreting cells such as *JCHAIN*, *MZB1*, and *IGHA1*. These patterns are consistent with established features of SLE immunopathology, including B-cell dysregulation and type I interferon activity.[10-12] However, PCA identifies major sources of variation across the dataset rather than genes that differ between clinical groups. Therefore, these findings do not establish that these pathways are specifically altered during progression from SLE to LN, requiring subsequent differential gene expression analysis between the disease groups.

Cell-type annotation identified approximately 3,353 B cells within the integrated PBMC dataset, with approximately 3,340 classified as naive B cells by fine resolution SingleR annotation. Expression of canonical markers including *TCL1A* and *IGHD* supported the assignment of this population. Although *TCL1A* has been associated with B-cell survival and altered B-cell states in autoimmune disease, its expression alone does not demonstrate autoreactivity, pathogenic activity, or a causal role in LN.[12,13] Instead, the identification of this population provides a defined cellular compartment for investigating whether B-cell transcriptional programs differ across disease states. This is particularly relevant to the central

objective of the study: determining whether molecular changes within B cells distinguish patients with SLE from those with LN.

Several limitations should be considered when interpreting these findings. The cohort consisted of only nine biological samples, with three individuals per clinical group. Although the dataset contained more than 70,000 cells after quality control, cells from the same individual are not independent biological replicates; therefore, the effective sample size remains three patients per group. This limits statistical power and the generalizability of subsequent differential expression analyses. In addition, the SLE samples were obtained from GSE263931, whereas the HC and LN samples were obtained from GSE306378. Although Harmony was used to reduce sample-associated technical variation, dataset and clinical condition are partially confounded, meaning that residual technical effects cannot be completely distinguished from biological differences. The analysis is also limited to PBMCs and therefore does not directly capture the tissue-specific immune environment of the kidney. Finally, automated annotation provides an efficient initial classification of cell populations but may mask additional heterogeneity within the B-cell compartment.[14]

Despite these limitations, the current analysis provides the necessary cellular framework for the next stage of the study. The annotated B-cell population can be isolated and examined at higher resolution to determine whether additional transcriptionally distinct populations are present within the current naive B-cell classification. Re-clustering and marker-based analysis could identify naive, memory, activated, plasmablast, or other B-cell states and determine whether their relative representation differs across HC, SLE, and LN. Differential gene expression analysis would then compare HC versus SLE, SLE versus LN, and HC versus LN. The SLE to LN comparison will be particularly important for identifying transcriptional changes

associated with renal involvement, while the HC comparisons will establish whether these changes represent broader lupus-associated abnormalities. Examining the overlap and differences between these comparisons will further distinguish shared disease-associated genes from changes that are more specifically associated with LN.[15-17]

Pathway enrichment analysis of the resulting differentially expressed genes will provide additional biological context by determining whether altered B-cell genes converge on processes such as type I interferon signaling, B-cell activation, antigen presentation, or inflammatory signaling.[18-20] Candidate genes and pathways identified through these analyses should subsequently be evaluated in larger, independently generated cohorts and, where possible, longitudinal or kidney-derived samples to determine their reproducibility and relevance to LN development. Longitudinal data would be particularly valuable for determining whether identified transcriptional changes precede renal involvement rather than simply reflecting established disease.[20,21]

Overall, this study successfully established a quality-controlled, integrated representation of the peripheral immune landscape and identified a defined B-cell population suitable for further investigation. Although the current results do not yet identify a specific B-cell transcriptional signature associated with LN, they provide the cellular and analytical foundation required to determine which genes and biological pathways distinguish SLE from SLE with renal involvement. These subsequent analyses may ultimately identify candidate molecular signatures for further validation and contribute to a better understanding of the biological changes associated with lupus nephritis.

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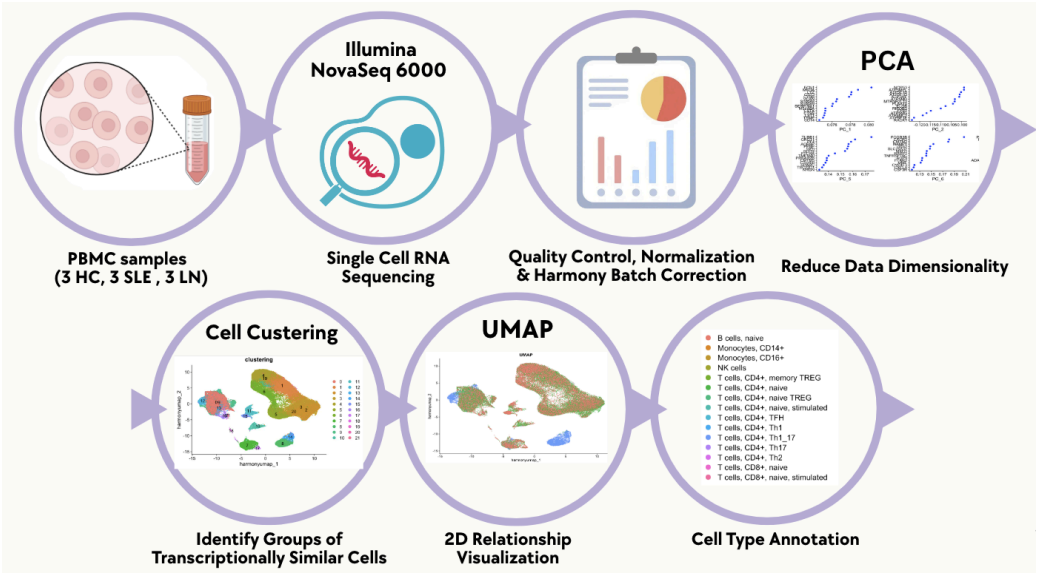
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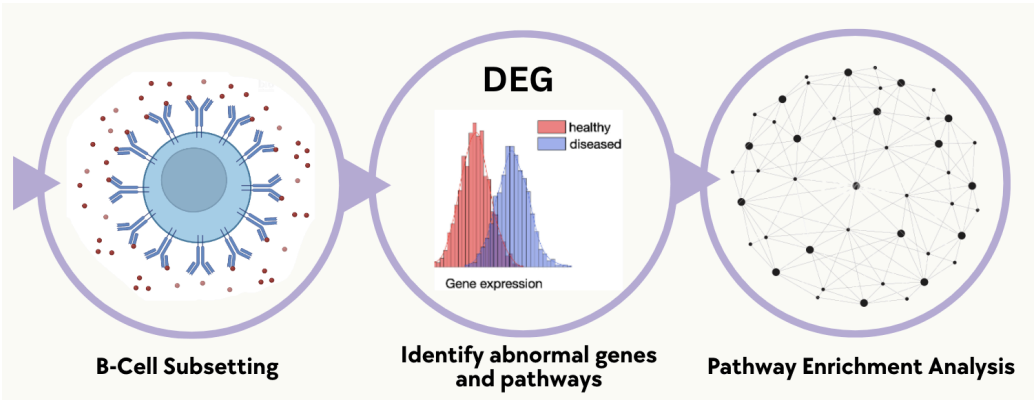
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Appendix



(A)



(B)

Figure 1. *Overview of the single-cell RNA sequencing analysis workflow. (A) Computational processing and analysis performed in this study, including quality control, normalization, Harmony integration, dimensionality reduction, cell clustering, UMAP visualization, and cell-type annotation. (B) Proposed downstream analyses, including B-cell subsetting, differential gene expression, and pathway enrichment analysis, to identify transcriptional signatures associated with SLE and lupus nephritis.*

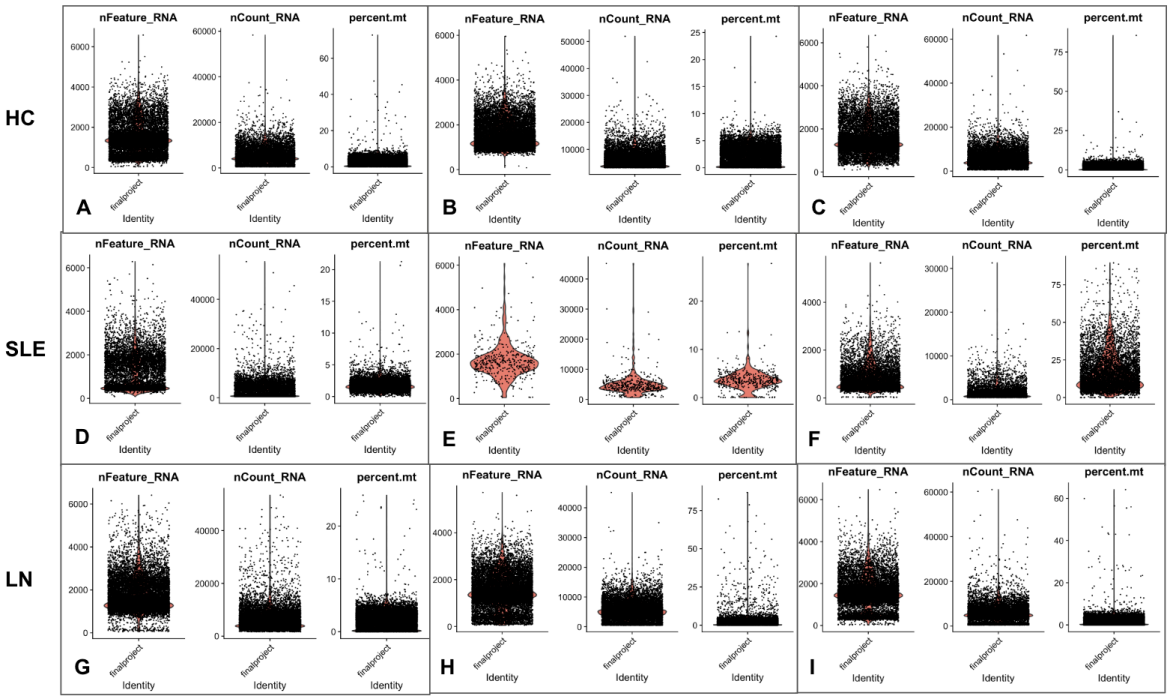


Figure 2. *Quality control of single-cell RNA sequencing data. Violin plots showing the distribution of the number of detected genes per cell (nFeature_RNA), total RNA counts per cell (nCount_RNA), and the percentage of mitochondrial transcripts (percent.mt) across experimental conditions. Quality control metrics were used to identify and remove low-quality cells and technical outliers prior to downstream analysis. (A-C) Healthy control (HC) samples, (D-F) SLE pre-treatment samples, and (G-I) show lupus nephritis (LN) samples.*

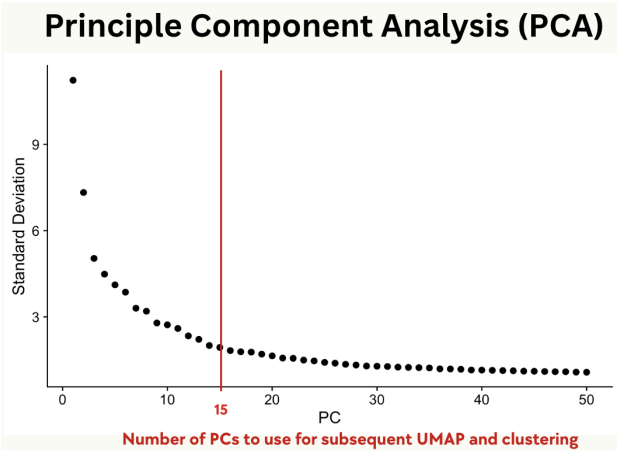


Figure 3. *Principal component analysis of single-cell transcriptomic data. Elbow plot showing the variance captured by the first 50 principal components. Principal components were ranked according to the amount of variation explained across cells, with the inflection point used to determine the number of dimensions retained for downstream clustering and UMAP visualization.*

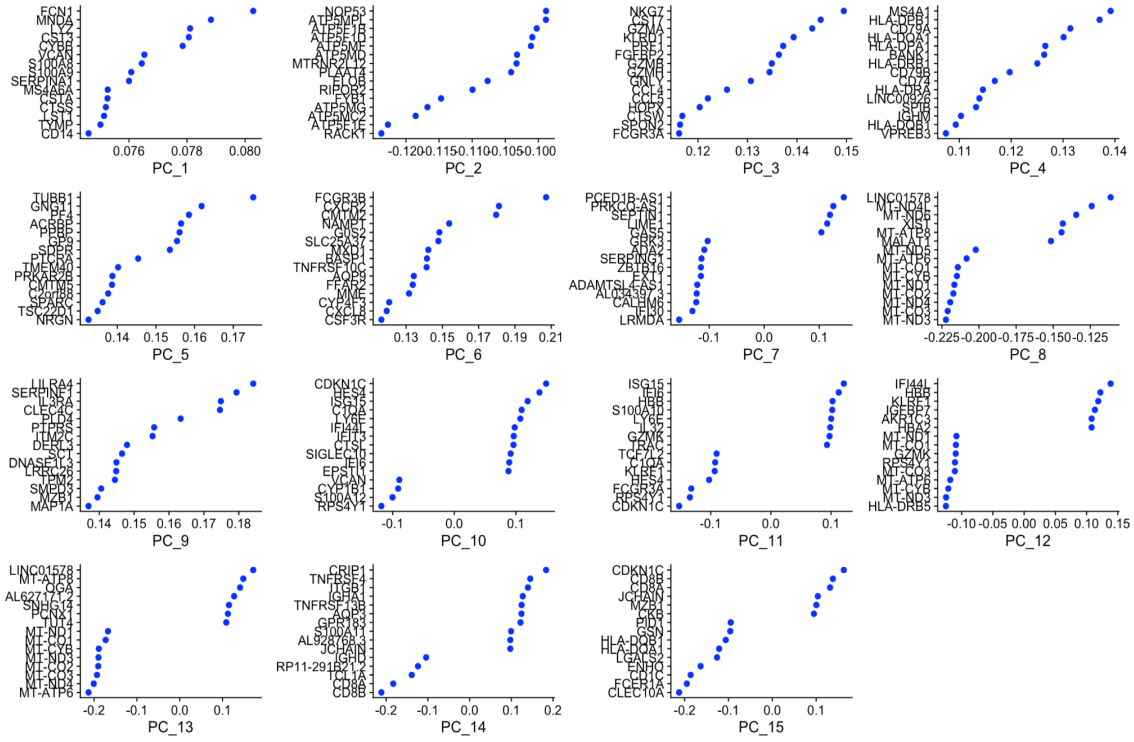


Figure 4. *Principal component analysis and gene contributions to transcriptional variation.*

Visualization of the first 15 principal components, showing the genes contributing most strongly to each component. Gene loading values indicate the relative contribution of each gene to the transcriptional variation captured by the corresponding principal component.

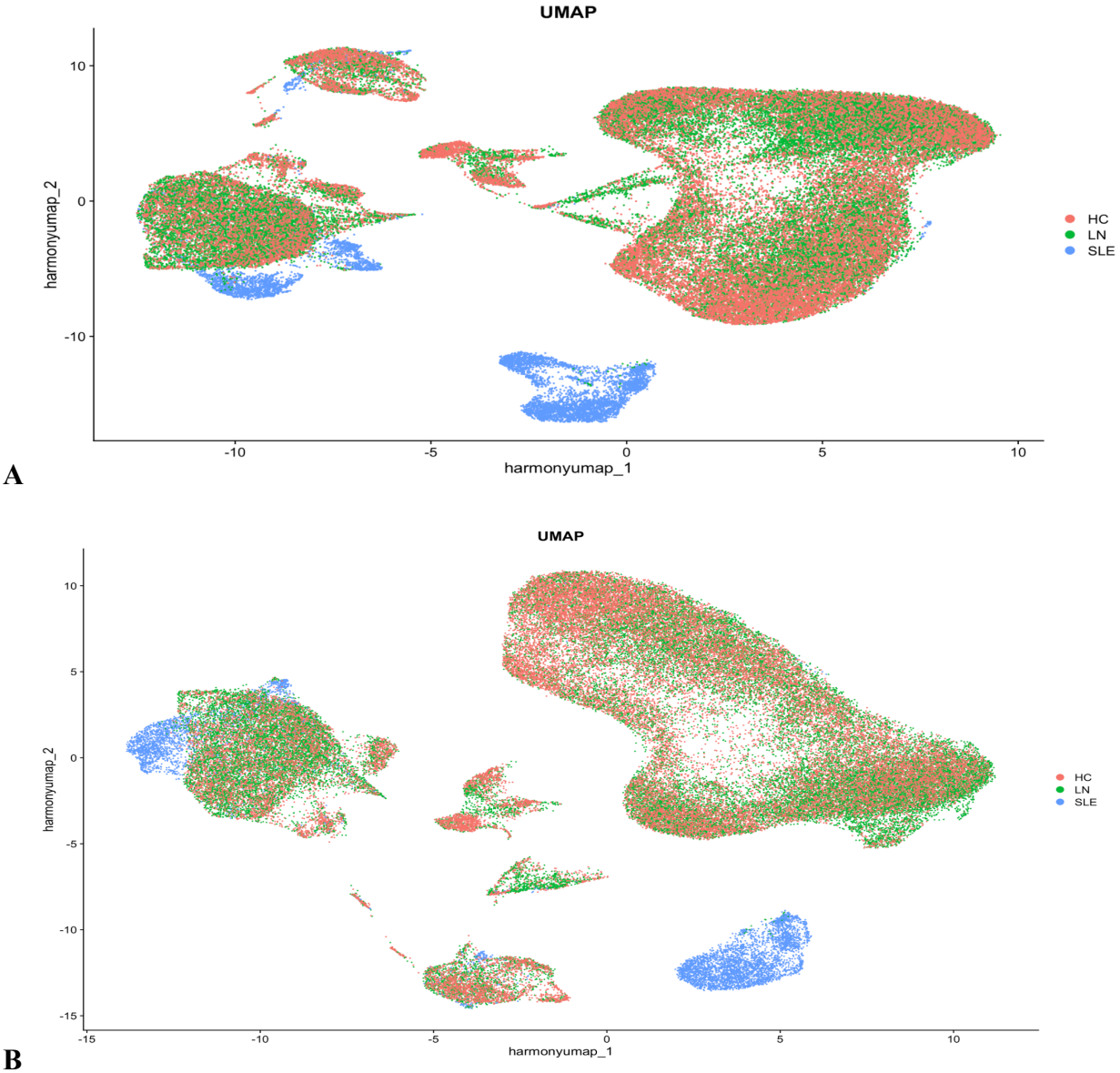
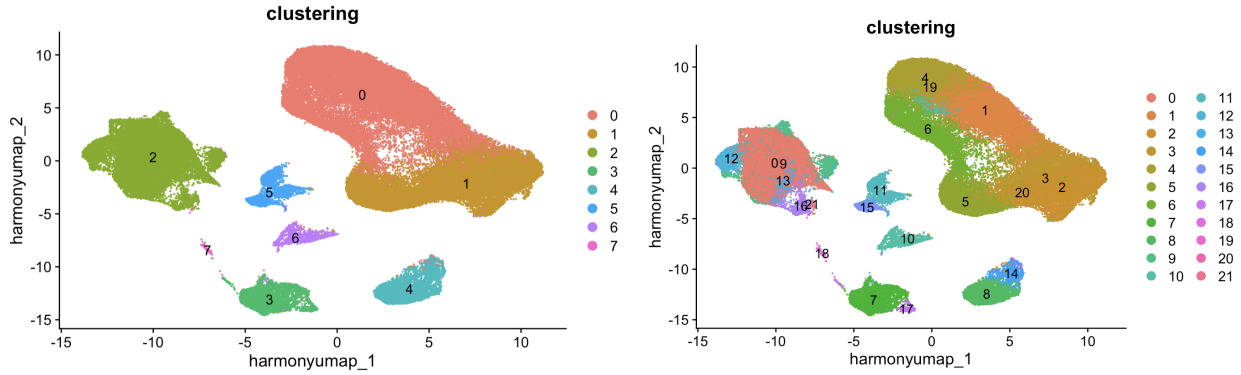


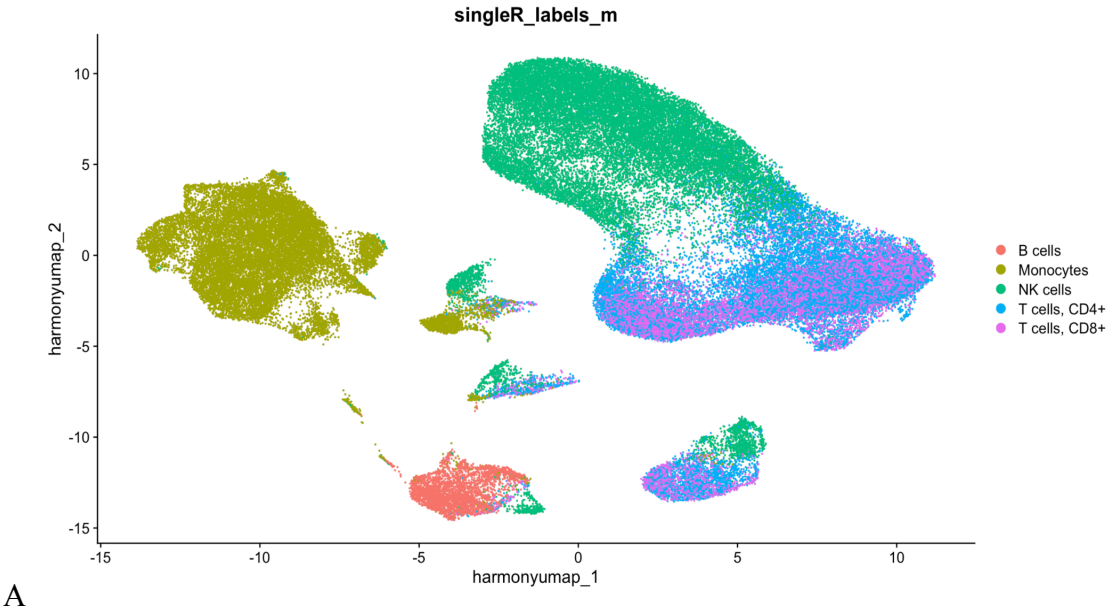
Figure 5. *UMAP visualization of the Harmony-integrated single-cell RNA sequencing dataset. UMAP projection generated from the Harmony-corrected principal component space following integration of the nine PBMC samples. (A) Cells colored according to individual sample identity,*

showing the distribution of cells from each sample within the integrated transcriptional space. (B) Cells colored according to clinical condition, including healthy controls (HC), systemic lupus erythematosus (SLE), and lupus nephritis (LN). Each point represents an individual cell, with spatial proximity reflecting similarity in transcriptional profiles after correction for sample-associated technical variation.

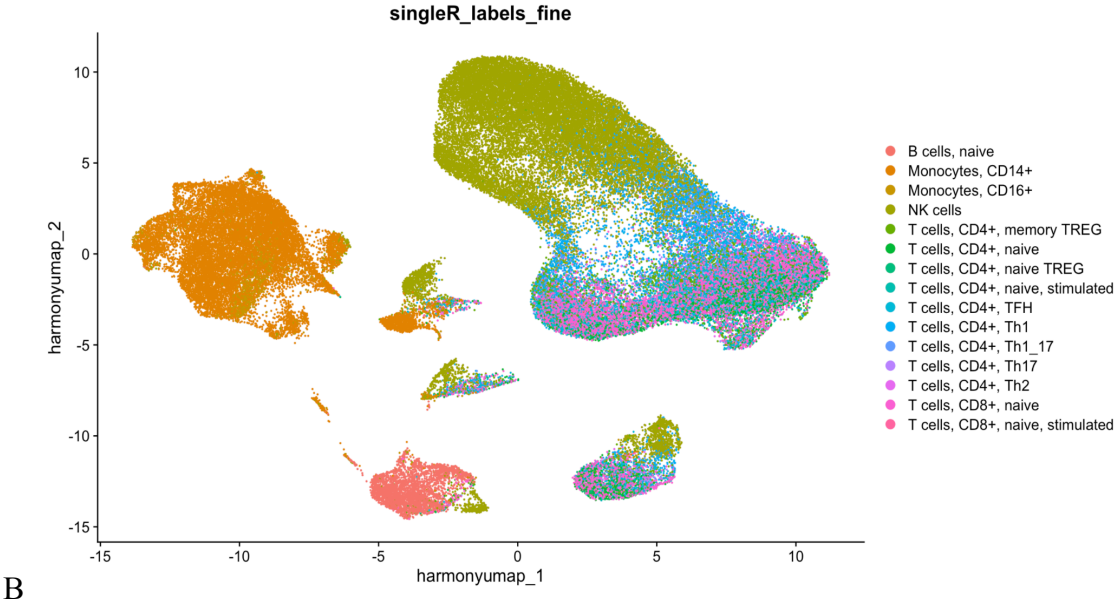


A **B**

Figure 6. *Graph-based clustering of the integrated single-cell RNA sequencing dataset. UMAP representation of Harmony-corrected single-cell transcriptomes following graph-based clustering. (A) Clustering at resolution 0.1. (B) Clustering at resolution 0.5. Numbers denote computationally defined clusters based on transcriptional similarity; each point represents an individual cell.*



A



B

Figure 7. *Automated cell-type annotation of the integrated PBMC dataset using SingleR. UMAP projections of the Harmony-corrected single-cell transcriptomes annotated using the celldex immune reference dataset.* ***(A)*** *Broad-resolution SingleR annotation (SingleR main labels) identifying major immune cell populations, including B cells, monocytes, NK cells, and T-cell populations.* ***(B)*** *Fine-resolution SingleR annotation (SingleR fine labels) providing more*

detailed classification of immune populations, including naive B cells and distinct T-cell and monocyte populations. Each point represents an individual cell, with colors indicating the assigned cell type.